Effect of a Novel Pharmacological Treatment on Nightmare Frequency in Patients with PTSD

A Paired-Sample Inferential Statistics Case Study

Prepared as part of an Inferential Statistics case-study series | Analyses performed in Python (SciPy, statsmodels) and R (base stats, pwr)

Abstract

Background: Chronic nightmares are a common and distressing symptom of PTSD. This case study evaluates whether an 8-week course of a novel drug reduces weekly nightmare frequency. **Methods:** Twelve patients were assessed for nightmare frequency (episodes/week) at baseline and after 8 weeks of treatment in a single-arm, pre/post paired design. A one-tailed Wilcoxon signed-rank test was pre-specified as the primary analysis, given the small sample size, the discrete/count nature of the outcome, and an observed outlier; a one-tailed paired t-test was reported as a secondary comparison. **Results:** All 12 patients showed a reduction in nightmare frequency. The Wilcoxon signed-rank test found the reduction statistically significant ($V = 78$, $p = 0.0012$), with a Hodges-Lehmann estimated median reduction of 3.5 nightmares/week (95% CI: at least 2.0). The paired t-test agreed ($t = 5.63$, $p < 0.001$, Cohen's $d = 1.63$), and achieved power at this effect size was >0.99. **Conclusion:** The observed reduction in nightmare frequency is unlikely to be due to chance, though the absence of a control arm and the small sample size limit causal interpretation.

1. Introduction / Background

Recurrent trauma-related nightmares are one of the more treatment-resistant symptoms of post-traumatic stress disorder (PTSD), and are a common target in pharmacological PTSD trials (e.g., prazosin studies for nightmare reduction). This case study simulates a small pilot evaluation of a new drug intended to reduce nightmare frequency, and works through the full inferential statistics workflow: hypothesis formulation, test selection, assumption checking, estimation, and honest reporting of limitations.

2. Methods

2.1 Study Design

Single-arm, pre/post paired design. Twelve patients with chronic PTSD-related nightmares had weekly nightmare frequency recorded at baseline and again after 8 weeks on the study drug. Because the same patients are measured twice, the before and after observations are dependent, ruling out independent-samples test procedures.

2.2 Hypotheses

A one-tailed hypothesis was pre-specified, reflecting the directional expectation that the drug reduces (not merely changes) nightmare frequency:

H_0 : median(before – after) = 0 (the drug has no effect)

H_1 : median(before – after) > 0 (the drug reduces nightmare frequency)

2.3 Statistical Analysis Plan and Test Selection

The Wilcoxon signed-rank test (one-tailed) was designated the **primary** analysis, with the paired t-test (one-tailed) reported as a secondary, comparison analysis. This choice was made for design-level reasons rather than purely as a reaction to the Shapiro-Wilk result on this sample: the outcome is a small-integer count (nightmares/week), a variable type that is characteristically right-skewed; the sample size (n=12) is small; tied difference values are present in the data (which by themselves preclude an exact Wilcoxon null distribution and point to coarse, non-continuous measurement); and one participant's values were visibly distant from the rest (an outlier that a mean-based test is more sensitive to than a rank-based one). We note explicitly that using a preliminary normality test's p-value to decide, post hoc, which subsequent test to run is a two-stage testing procedure that can itself distort the nominal Type I error rate; accordingly, the rank-based test was treated as primary by design rather than by a normality-test cutoff alone.

3. Results

3.1 Raw Data

Patient	Before (nightmares/wk)	After (nightmares/wk)	Difference
Patient_1	7	2	5
Patient_2	5	3	2
Patient_3	9	4	5
Patient_4	4	3	1
Patient_5	8	5	3
Patient_6	12	6	6
Patient_7	6	4	2
Patient_8	5	4	1
Patient_9	10	3	7
Patient_10	7	5	2
Patient_11	6	4	2
Patient_12	15	9	6

3.2 Descriptive Statistics

Metric	Before	After	Difference
Mean	7.83	4.33	3.50
SD	3.21	1.83	2.15
Median	7.00	4.00	2.50

3.3 Diagnostic Plots and Normality Assessment

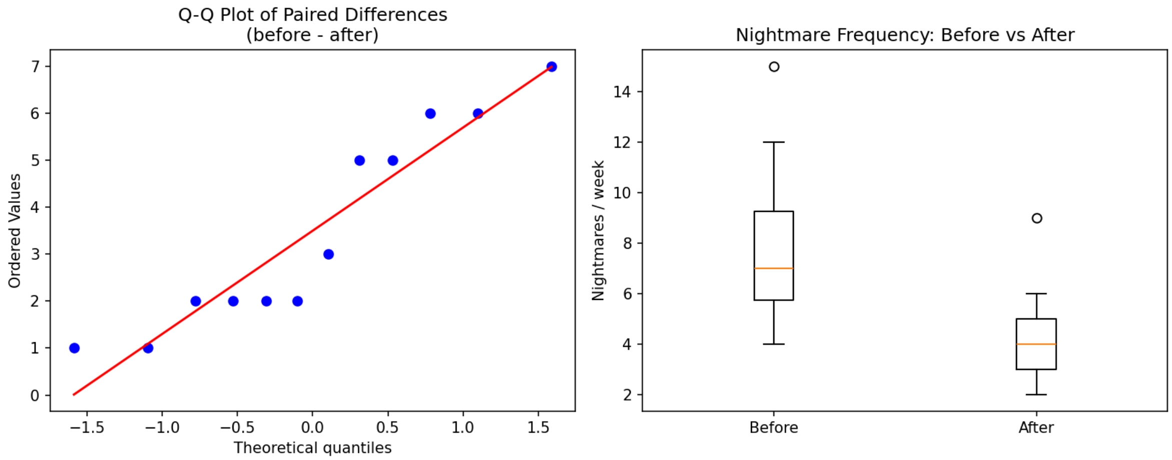

Figure 1. Left: Q-Q plot of paired differences against a normal reference line. Right: boxplots of raw before/after values, with Patient_12 flagged as an outlier in both conditions.

The Shapiro-Wilk test on the paired differences was not statistically significant ($W = 0.872$, $p = 0.070$). Given n=12, this test has limited power to detect non-normality, so the result is treated as inconclusive rather than confirmatory. The Q-Q plot shows mild deviation from the reference line in the upper tail, consistent with the outlier visible in the boxplot.

3.4 Primary Analysis: Wilcoxon Signed-Rank Test

The one-tailed Wilcoxon signed-rank test (with continuity correction, due to tied values) found a statistically significant reduction in nightmare frequency following treatment: **$V = 78$, $p = 0.0012$**. The Hodges-Lehmann estimate of the median reduction was **3.50 nightmares/week** (one-sided 95% CI: ≥ 2.00). The matched-pairs rank-biserial correlation, an effect-size measure for this test, was 1.00 — the maximum possible value, reflecting that all 12 patients improved.

3.5 Secondary Analysis: Paired t-test

As a comparison analysis, the one-tailed paired t-test also found a statistically significant reduction: **$t(11) = 5.63$, $p < 0.001$**. Cohen's d was 1.63, conventionally interpreted as a very large effect. The two tests agree in both direction and significance.

3.6 Power Analysis

A post-hoc power calculation at the observed effect size ($d = 1.63$, $n = 12$, $\alpha = 0.05$, one-tailed) indicated achieved power of approximately 0.9998. An a priori calculation indicated that only 5 patients would have been required to detect an effect of this size at 80% power. This is reported as a caution, not a strength: the specific effect observed in this sample happened to be large enough to detect even with n=12, which does not imply that n=12 is generally an adequate sample size for a trial of this kind. A pre-registered a priori power calculation, based on a clinically meaningful minimum effect size rather than the observed one, would be expected before a real confirmatory trial.

4. Discussion

All 12 patients in this sample showed a reduction in nightmare frequency after 8 weeks of treatment, with the median falling from 7 to 4 episodes per week. Both the primary (Wilcoxon) and secondary (paired t-test) analyses indicate this reduction is unlikely to be attributable to chance. The consistency of direction across all patients (12/12 improved) and the agreement between a rank-based and a mean-based test strengthen confidence in the finding, despite the small sample.

That said, a statistically significant pre/post difference in a single-arm design is not, by itself, evidence that the drug caused the improvement. Nightmare frequency in PTSD can fluctuate over time independent of treatment, and simply enrolling in a study and receiving regular clinical attention can itself produce improvement (a placebo or Hawthorne-type effect). A randomized, placebo-controlled design would be required to isolate the drug's specific contribution.

5. Limitations

- **No control or placebo arm** — the design cannot rule out natural symptom fluctuation, regression to the mean, or placebo response as alternative explanations for the observed reduction.
- **Small sample size (n=12)** — individual patients, particularly the outlier (Patient_12), can meaningfully influence summary statistics; estimates should be treated as imprecise.
- **Self-reported outcome** — nightmare counts were patient-reported, introducing potential recall bias and expectation effects.
- **Tied values in the differences** — several patients shared identical difference values, which technically precludes computation of an exact (rather than continuity-corrected, asymptotic) Wilcoxon p-value.
- **Post-hoc power framing** — the high achieved power reflects the specific effect size observed in this sample and should not be interpreted as evidence that n=12 is generally sufficient for studies of this kind.

6. Conclusion

In this 12-patient pre/post case study, 8 weeks of the study drug was associated with a statistically significant and consistent reduction in weekly nightmare frequency (Hodges-Lehmann estimated reduction: 3.5 episodes/week; Wilcoxon $p = 0.0012$; paired t-test $p < 0.001$). These results support proceeding to a larger, randomized, placebo-controlled trial to establish whether the drug causes the observed improvement, rather than being treated as confirmatory evidence on their own.

7. Software and Reproducibility

All analyses were performed independently in both Python (SciPy 1.17, statsmodels 0.15) and R (base stats, pwr package), with results cross-checked for agreement. Full analysis code is included alongside this report (`ptsd_nightmare_case_study_v2.py` and `ptsd_nightmare_case_study_v2.R`).

References

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